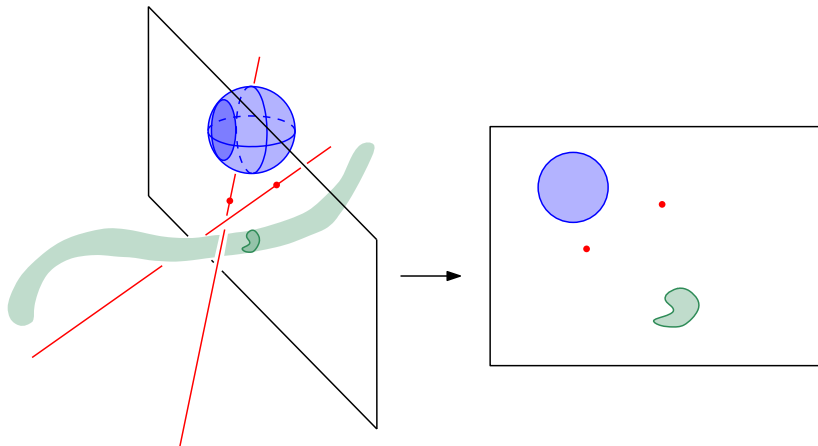


Stiefel Manifolds and Mass Partition Results.

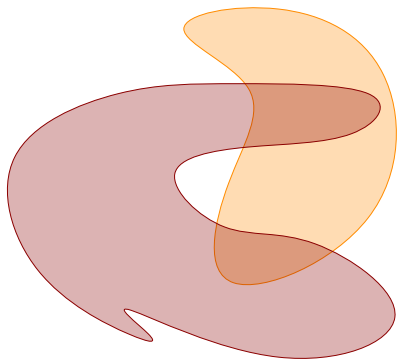


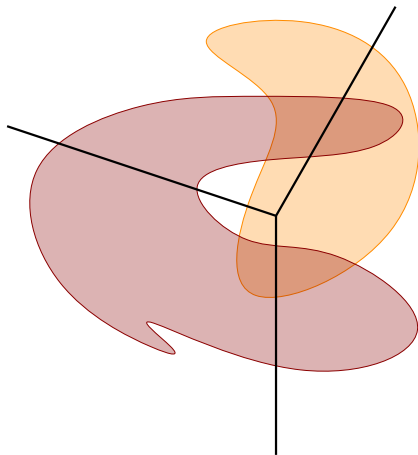
Pablo Soberón
Baruch College, City University of New York

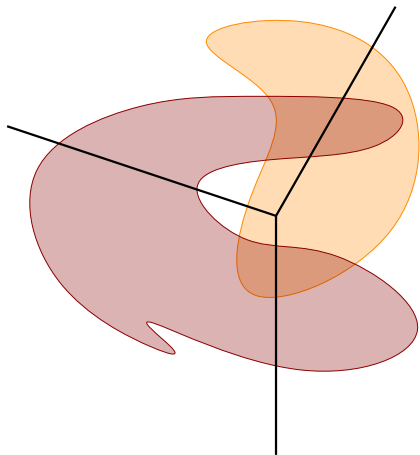
Collaborators

- ▶ Ilani Axelrod-Freed (MIT)
- ▶ Michael. N. Manta (Clatech)
- ▶ Yuki Takahashi (Grinnell College)

MASS PARTITION PROBLEMS

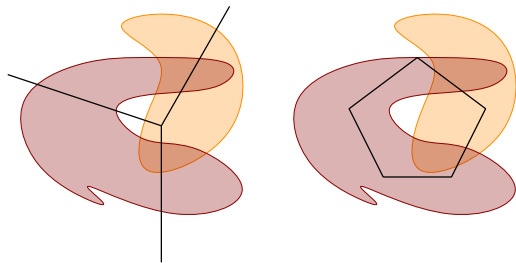






Question (Mass Partition problem)

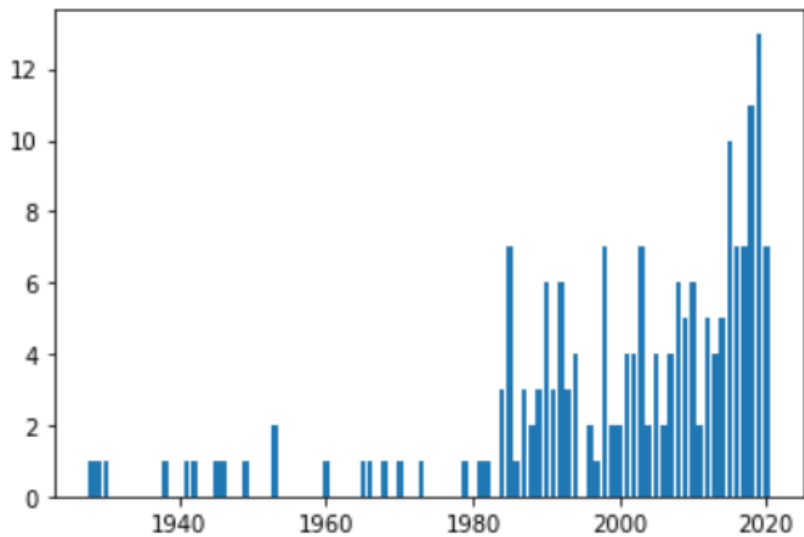
*If I give you **geometric constraints** on how to split $\mathbb{R}^d$, how can you split a family of measures?*



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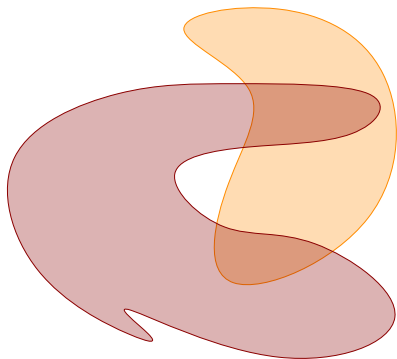
- ▶ Continuous measures vs Discrete measures
- ▶ Convex partitions, polynomial partitions
- ▶ Algorithmic results vs existence results
- ▶ High dimensional results

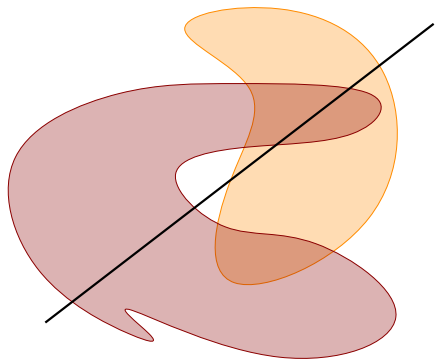


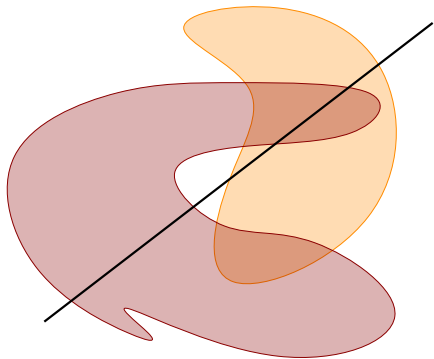
Message of the talk

Stiefel manifolds are useful to solve mass partition problems and to **simplify** proofs of existing results.

HISTORY AND FIRST EXAMPLES

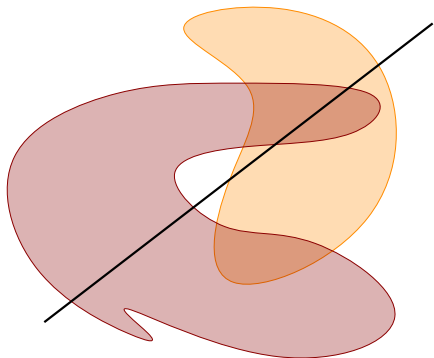






Theorem (Ham sandwich theorem, Banach 1938)

Given d measures in $\mathbb{R}^d$ there exists a hyperplane simultaneously halving all of each of them.

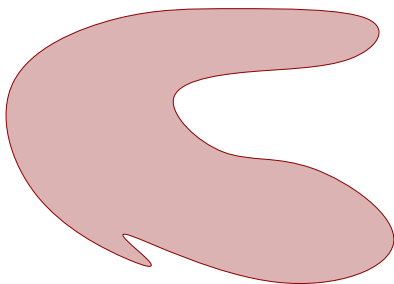


Theorem (Ham sandwich theorem, Banach 1938)

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Corollary

For any measure in $\mathbb{R}^2$ there are *two lines* halving it into *four equal parts*.

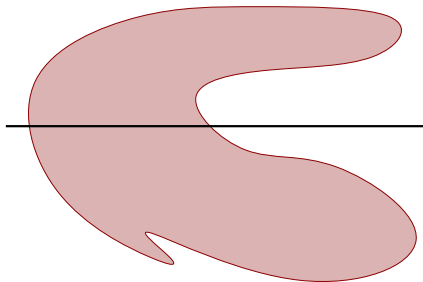


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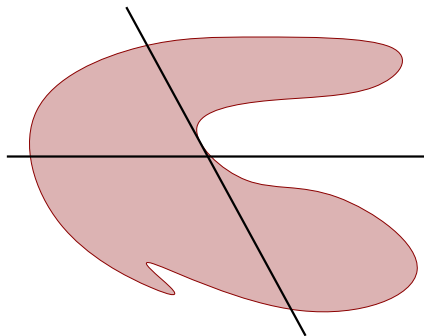


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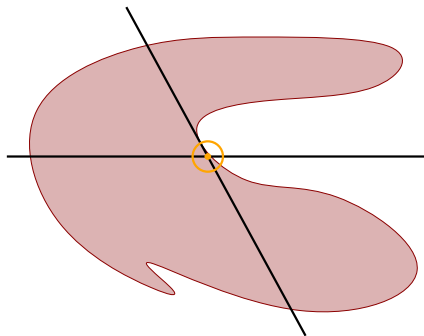


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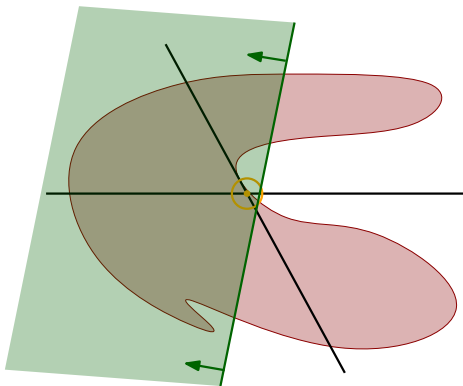


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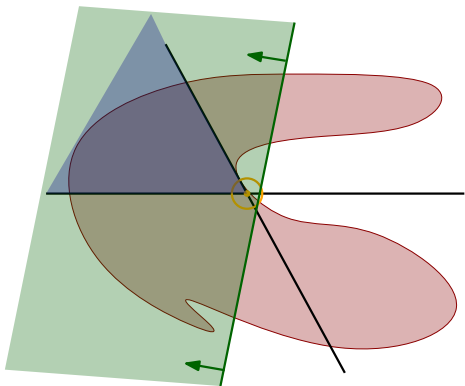


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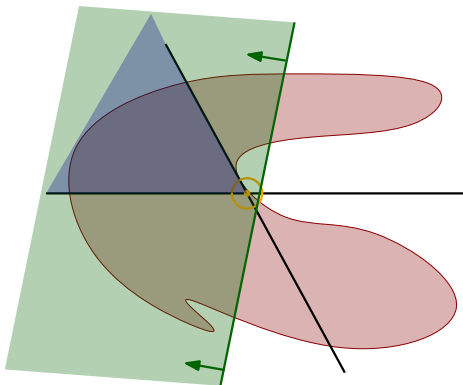


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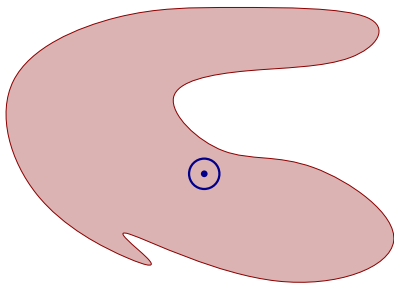
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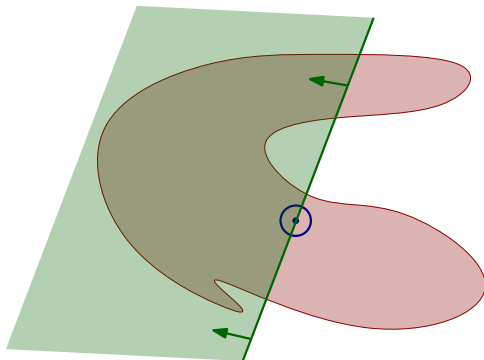
- Every half-space containing the intersection of these two lines has at least $1/4$ of the measure.



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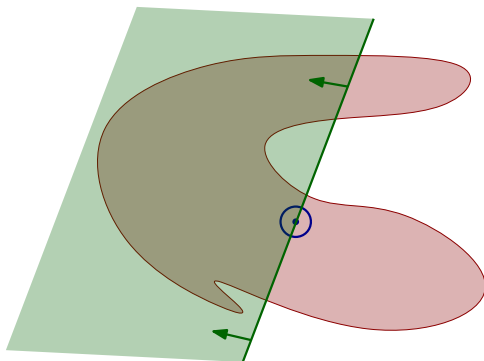
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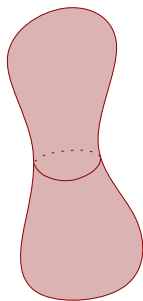
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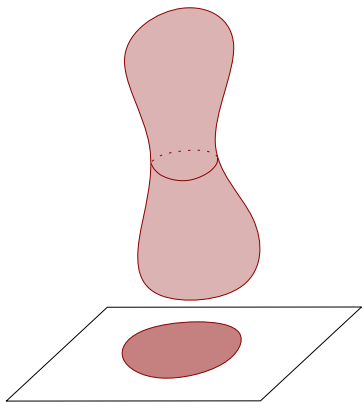
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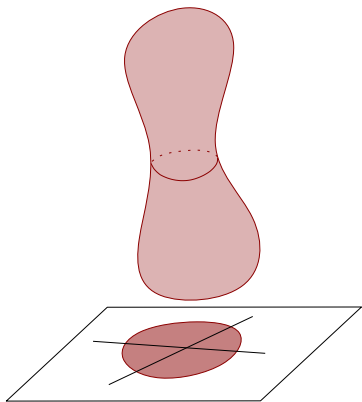


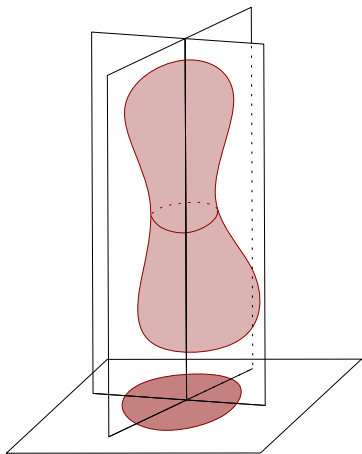
Theorem (Centerpoint theorem, Rado 1946)

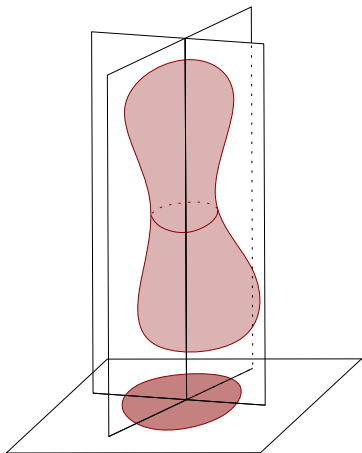
Given a measure in $\mathbb{R}^d$, there exists a point p such that any half-space that contains it has at least $1/(d+1)$ of the measure.





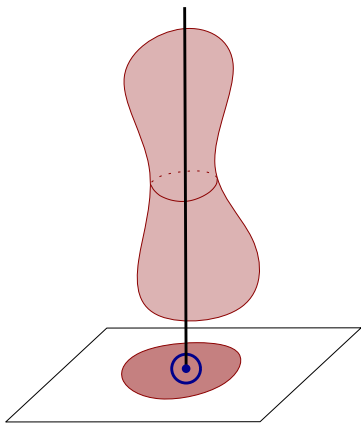
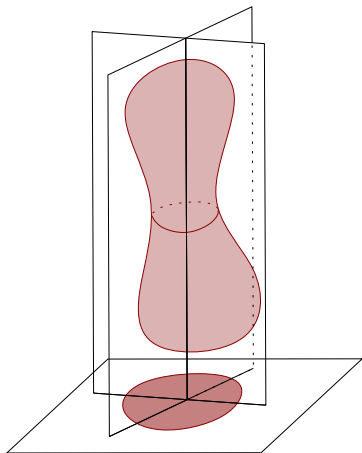






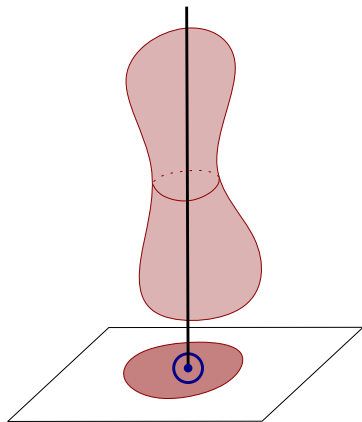
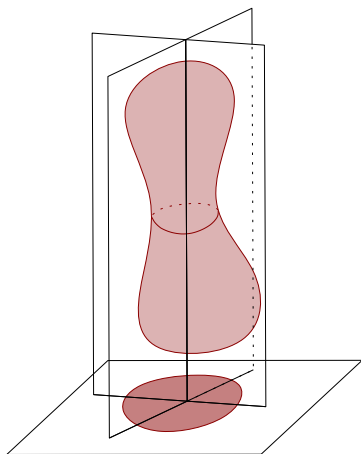
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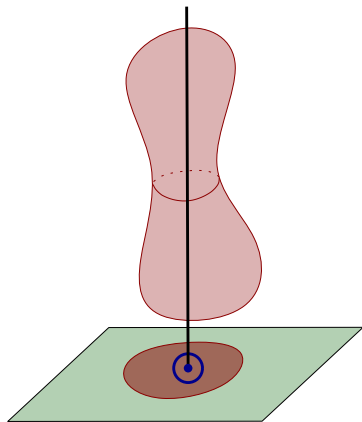
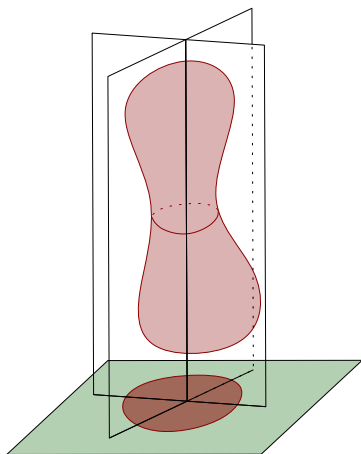


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For any measure in $\mathbb{R}^3$ there are two planes that cut it into four equal pieces

Corollary (centerline)

For any measure in $\mathbb{R}^3$ there exists a *line* such that any half-space that contains it has at least $1/3$ of the measure

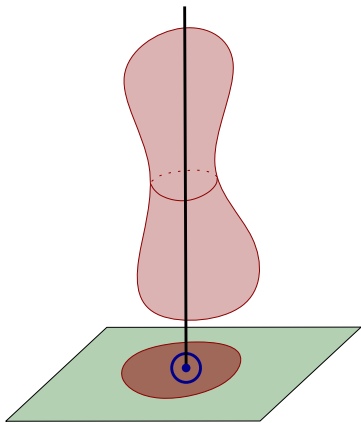
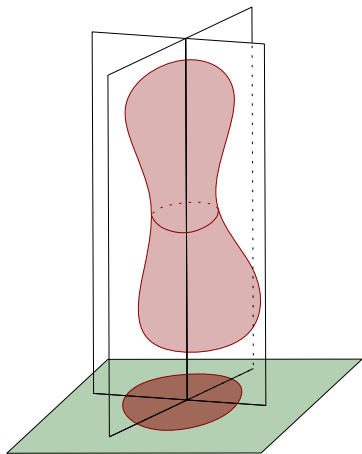


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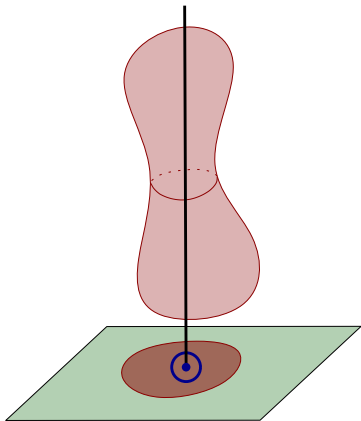
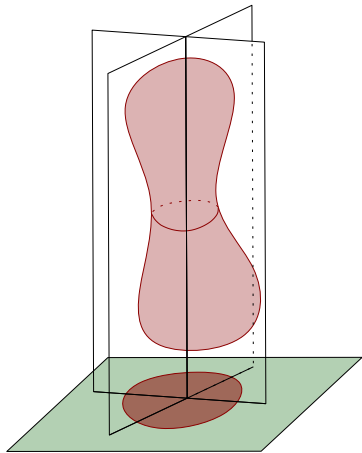
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Theorem (Hadwiger 1960)

For any **two** measures in $\mathbb{R}^3$ there are two planes that cut each into four equal pieces

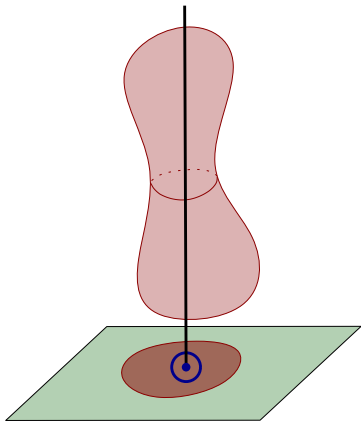
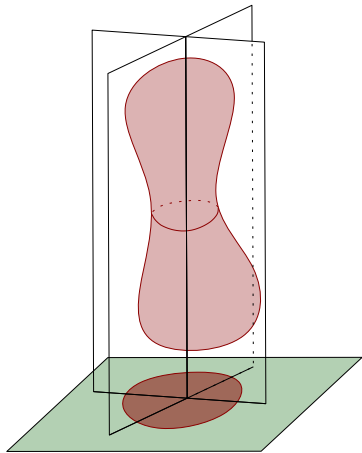


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Theorem (Živaljević, Vrećica 1990, Dolnikov 1992)

For any *two* measures in $\mathbb{R}^3$ there exists a common centerline

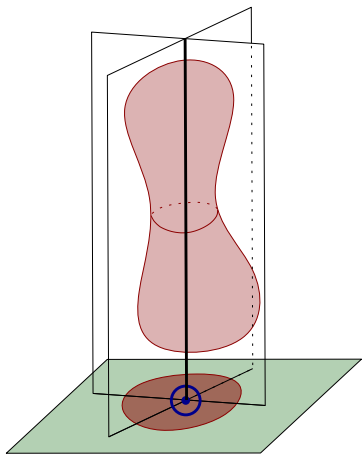


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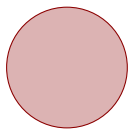
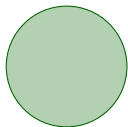
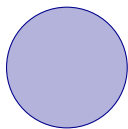
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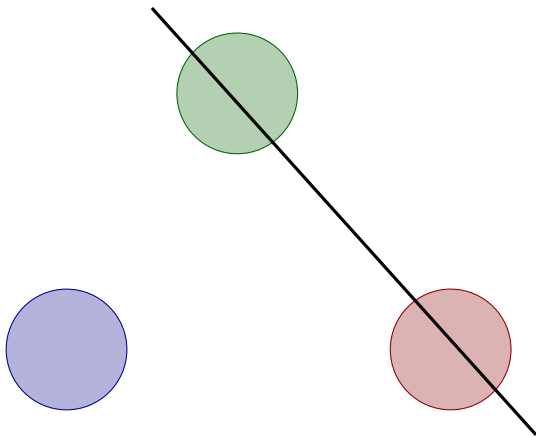
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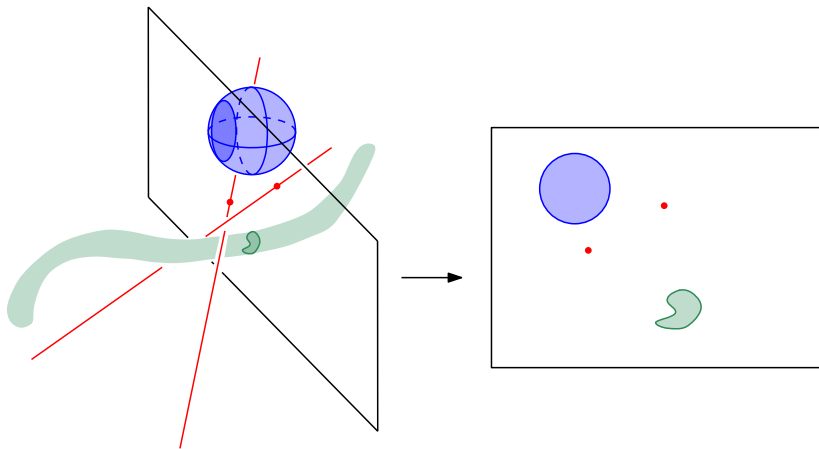


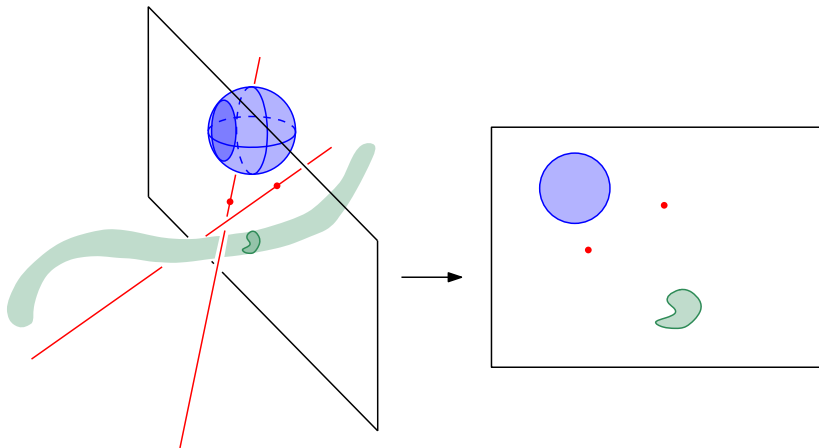
Theorem (Manta, Soberón 2021+)

For any *two* measures μ_1, μ_2 in $\mathbb{R}^3$ there exists two planes that split μ_1 into four equal parts whose intersection is a centerline for μ_2 .



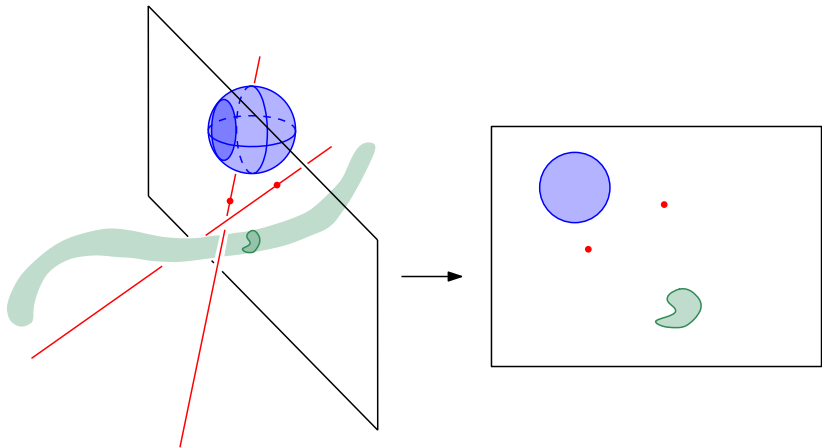






Question

Can we choose a plane in $\mathbb{R}^3$ so that the *three measures* induced can be simultaneously halved by a line?

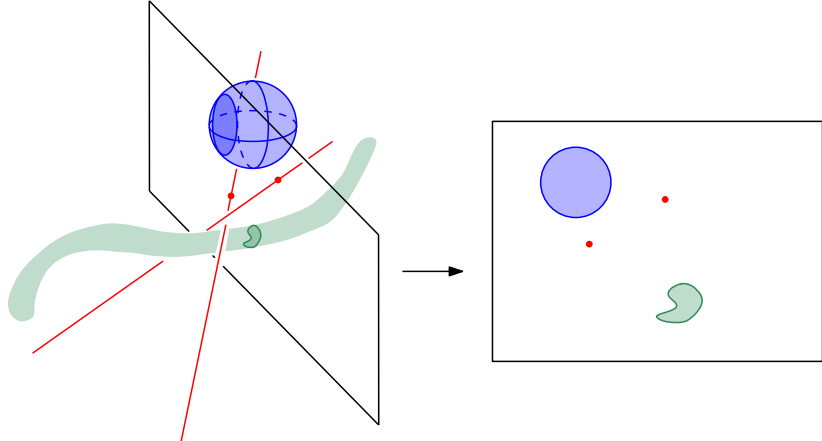


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Definition

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Theorem (Schnider 2020)

For any d k -dimensional mass assignments in $\mathbb{R}^d$, there is a k -dimensional affine subspace in which the induced measures can be halved simultaneously.

Example

Example

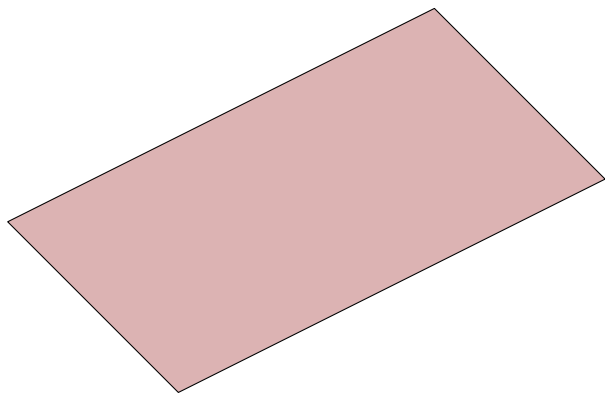
Suppose that in $\mathbb{R}^3$ we are given

- ▶ three measures,
- ▶ two 2-dimensional mass assignments, and
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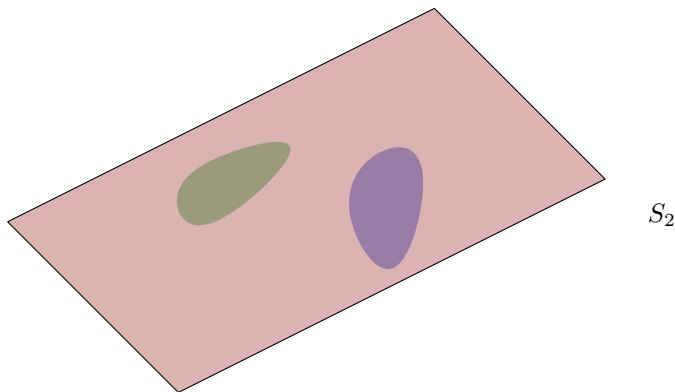


S_2

Example

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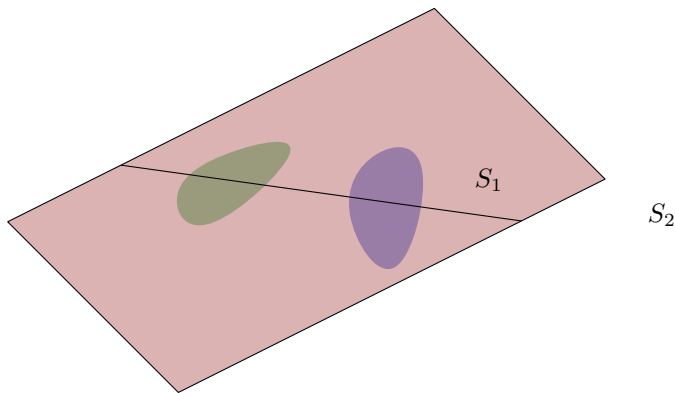
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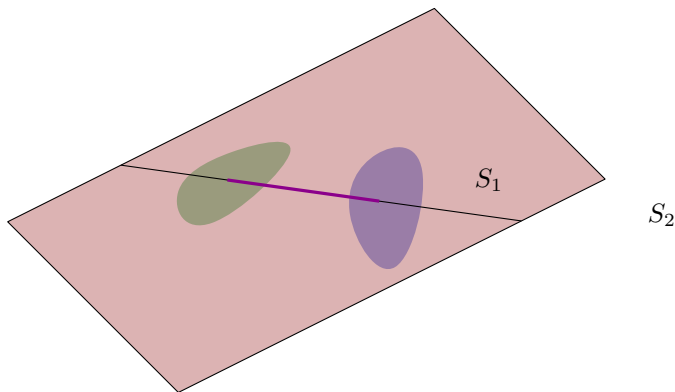
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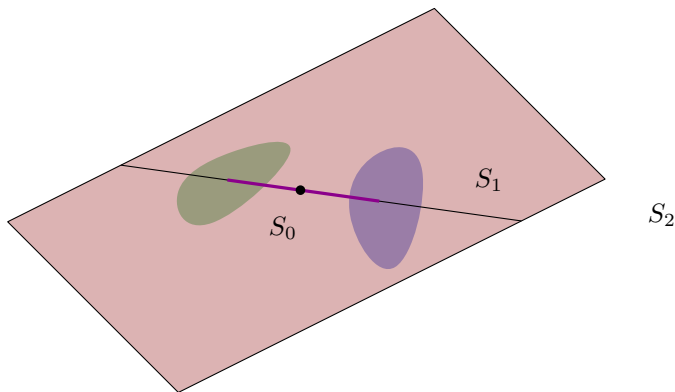
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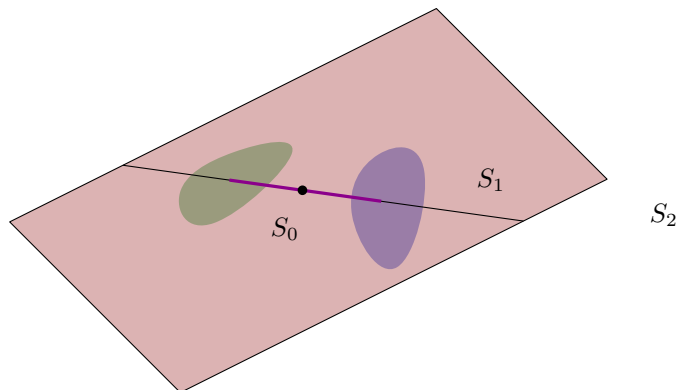
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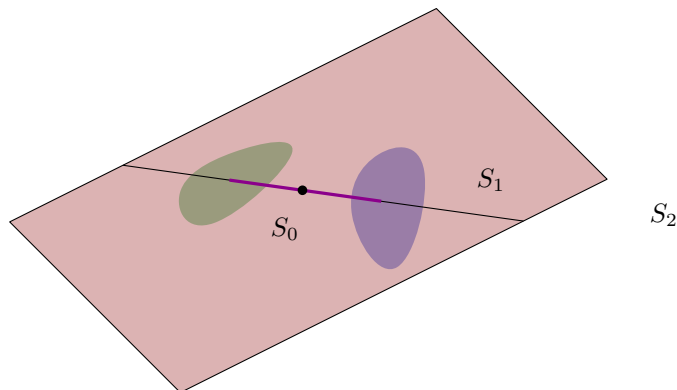


We get a flag $S_0 \subset S_1 \subset S_2 \subset S_3 = \mathbb{R}^3$ where S_i halves all the mass assignments on S_{i+1} .

Example

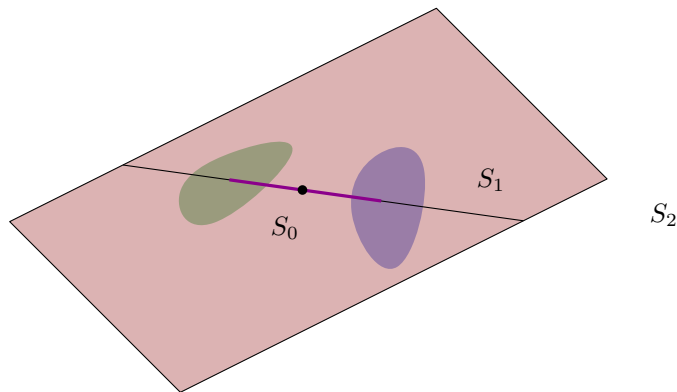
Suppose that in $\mathbb{R}^3$ we are given

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Do we have a flag $S_0 \subset S_1 \subset S_2 \subset S_3 = \mathbb{R}^3$ where S_i halves all the mass assignments on S_{i+1} ?

Example



Theorem (Axelrod-Freed, Soberón 2021+)

Let $k \leq d$ and $\pi = (\pi_d, \dots, \pi_k)$ be a permutation of $(d, \dots, k)$. For each i we are given π_i i -dimensional mass assignments in $\mathbb{R}^d$. Then, there exists a full flag $S_{k-1} \subset \dots \subset S_{d-1} \subset S_d = \mathbb{R}^d$ of affine spaces such that S_i halves all the π_{i+1} measures induced on S_{i+1} .

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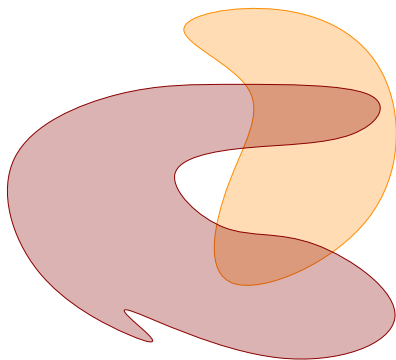
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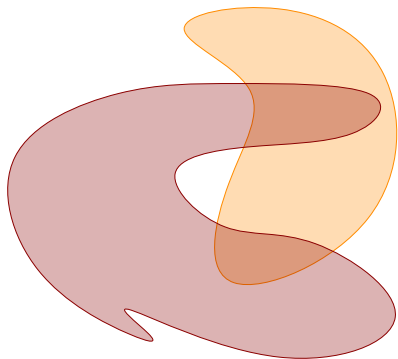
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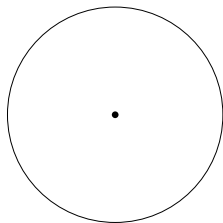
Consider $\pi = (d-1, d-2, \dots, k, d)$.

PROOF METHODS

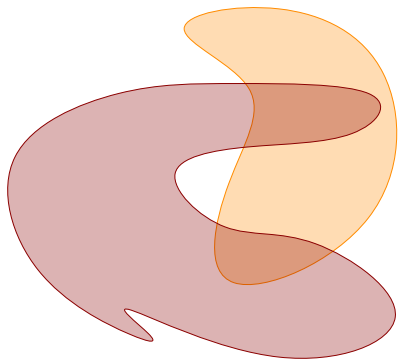




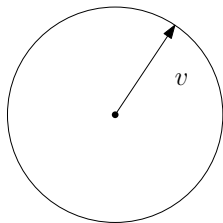
$\mathbb{R}^d$



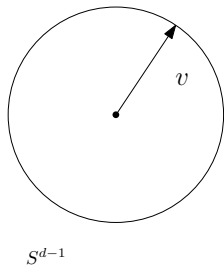
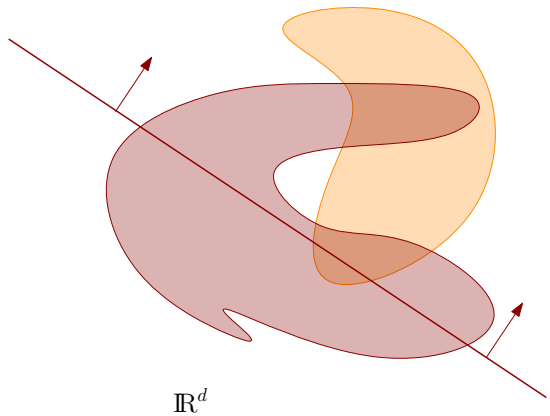
S^{d-1}

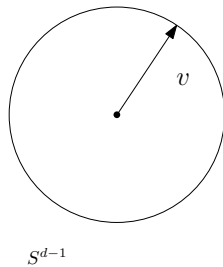
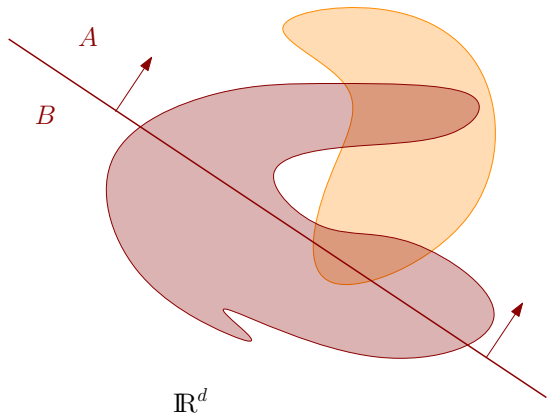


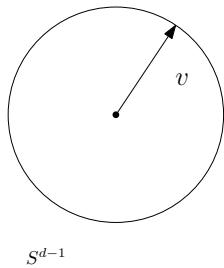
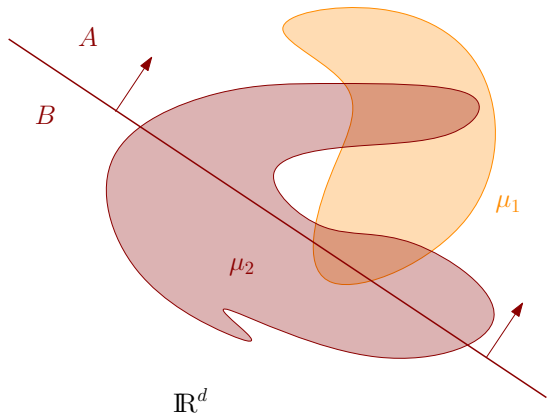
$\mathbb{R}^d$

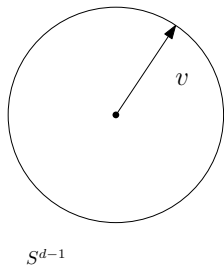
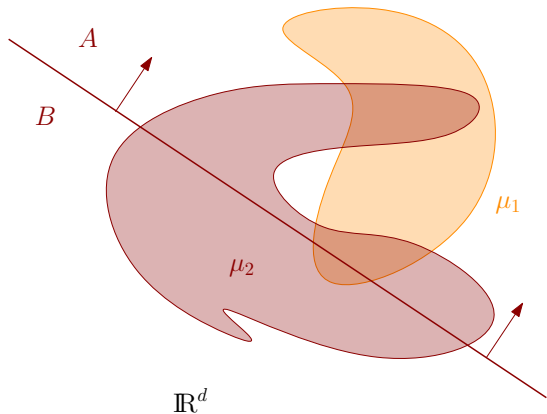


S^{d-1}





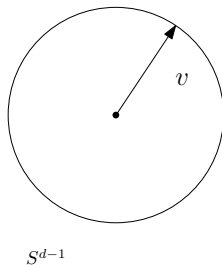
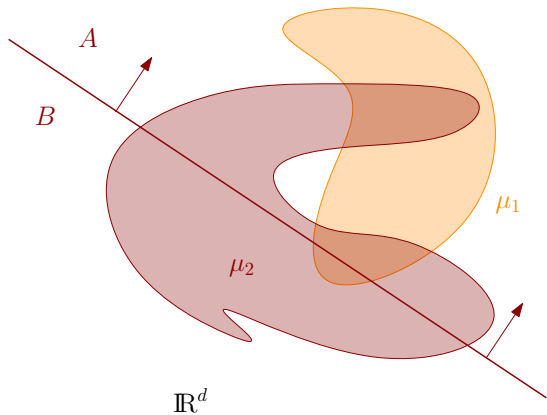




Consider

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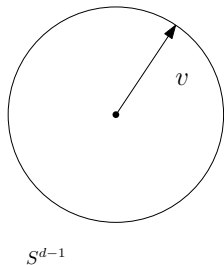
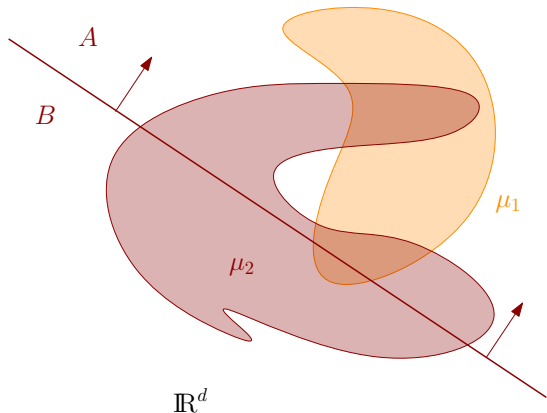


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Notice $f(-v) = -f(v)$.



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Theorem (Borsuk–Ulam)

For any continuous **odd** function $f : S^{d-1} \rightarrow \mathbb{R}^{d-1}$ has a zero.

Test map scheme

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- ▶ Ideally, general properties of equivariant maps $f : X \rightarrow_G Y$ will translate to useful statements for the mass partition problems.

Example

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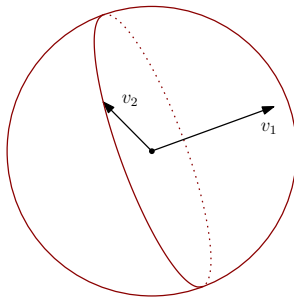
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$$V_k(\mathbb{R}^d) = \{(v_1, \dots, v_k) : v_1, \dots, v_k \text{ are orthonormal}\}.$$

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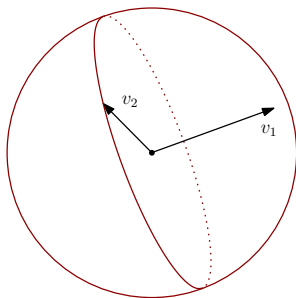
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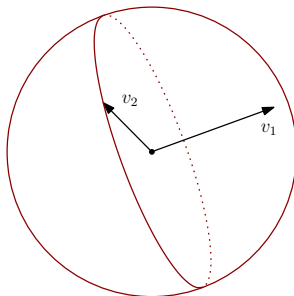


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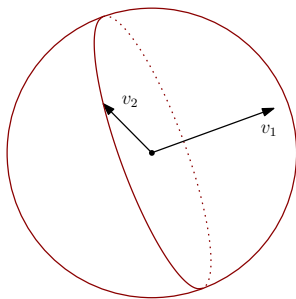
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$$\dim(V_k(\mathbb{R}^d)) = (d-1) + (d-2) + \dots + (d-k)$$

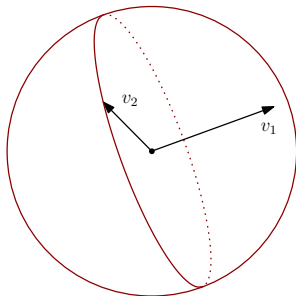
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- For $(v_1, \dots, v_k) \in V_k(\mathbb{R}^d)$ and $(\lambda_1, \dots, \lambda_k) \in (\mathbb{Z}_2)^k = \{-1, 1\}^k$ we define

$$(\lambda_1, \dots, \lambda_k) \cdot (v_1, \dots, v_k) = (\lambda_1 v_1, \dots, \lambda_k v_k) \in V_k(\mathbb{R}^d).$$

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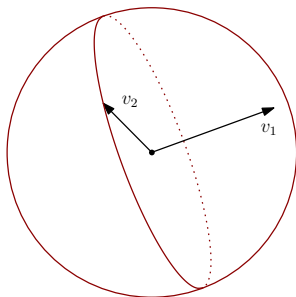


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- We have a similar action on $\mathbb{R}^{d-1} \times \dots \times \mathbb{R}^{d-k}$.

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Every continuous, $(\mathbb{Z}_2)^k$ -equivariant map
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THE FAIRY BREAD SANDWICH THEOREM

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Theorem (Axelrod-Freed, Soberón 2021+)

Let $1 \leq k \leq d$ and $\pi = (\pi_d, \dots, \pi_k)$ be a permutation of $(d, \dots, k)$. For each i we are given π_i i -dimensional mass assignments in $\mathbb{R}^d$. Then, there exists a full flag $S_{k-1} \subset \dots \subset S_{d-1} \subset S_d = \mathbb{R}^d$ of affine spaces such that S_i halves all the π_{i+1} measures induced on S_{i+1} .

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Proof.

- We take out one mass assignment of each dimension, $\mu_{d,0}, \dots, \mu_{k,0}$ and define $\tau_{i-1} = \pi_i - 1$. Then, $\tau = (\tau_{d-1}, \dots, \tau_{k-1})$ is a permutation of $(d-1, \dots, k-1)$.

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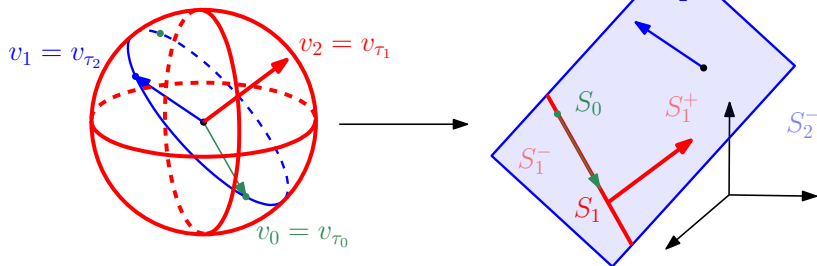
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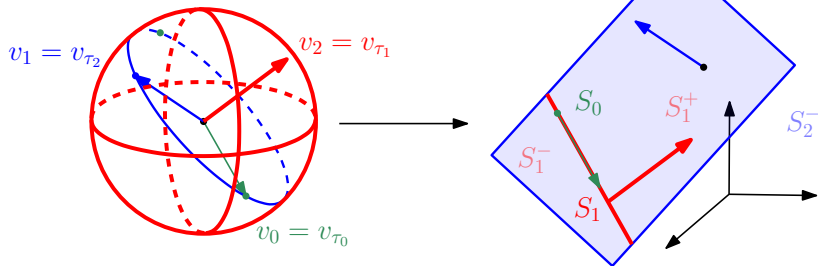
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$$\begin{aligned} f : V_{d-k+1}(\mathbb{R}^d) &\rightarrow \mathbb{R}^{d-1} \times \dots \times \mathbb{R}^{k-1} \\ (v_{d-1}, \dots, v_{k-1}) &\mapsto (x_{d-1}, \dots, x_{k-1}). \end{aligned}$$

where x_{τ_i} tells us how S_i splits each of the τ_i mass assignments on S_{i+1} for all i .

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We can impose additional geometric conditions on the k -dimensional subspaces we seek.

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Given d k -dimensional mass assignments in $\mathbb{R}^d$, there exists a k -dimensional linear space $S_k \subset \mathbb{R}^d$ containing rays in $k - 1$ fixed directions and a $(k - 1)$ -dimensional affine subspace $S_{k-1} \subset S_k$ that halves each of the measures induced on S_k .

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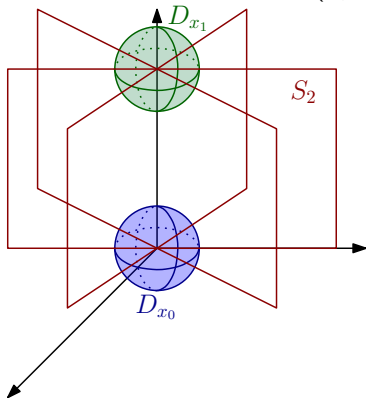
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CENTRAL TRANSVERSALS

Definition

Let $0 \leq \lambda < d$ be integers. Given a finite measure μ in $\mathbb{R}^d$, we say that an affine space L of dimension λ is a **λ -transversal** to μ if every half-space that contains L also has at least $\frac{1}{d-\lambda+1}$ of the measure μ .

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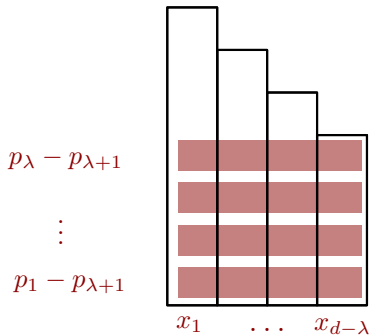
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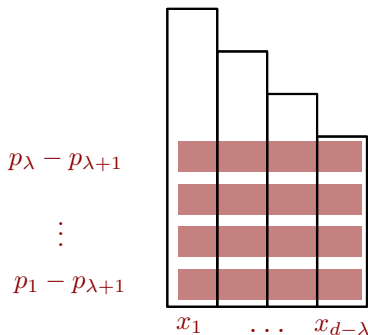
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For $1 \leq j \leq \lambda$, the j -th coordinate of x_i is $\langle v_i, p_j - p_{\lambda+1} \rangle$. The rest are zero.

Theorem (Manta, Soberón 2021+)

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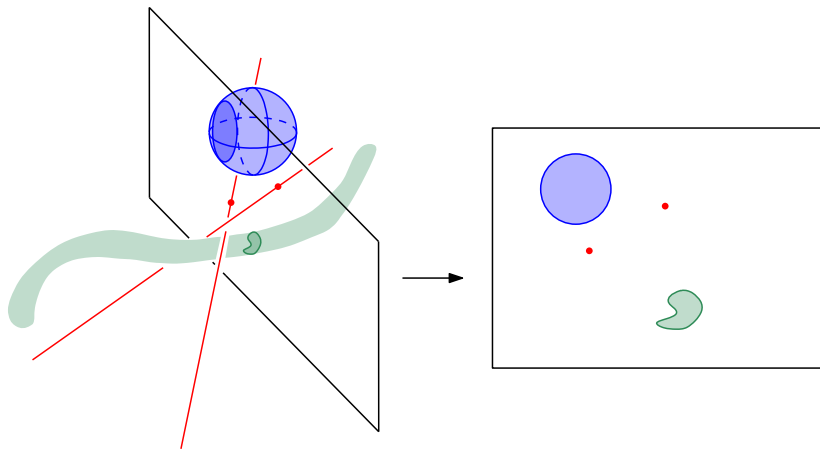
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Let $\mu_1, \dots, \mu_d$ be finite measures in $\mathbb{R}^d$. Then, there exist two hyperplane H_1, H_2 such that

- ▶ $H_1 \cup H_2$ splits μ_d into four equal parts,
- ▶ H_1 halves μ_{d-1} , and
- ▶ $H_1 \cap H_2$ is a $(d-2)$ -transversal for $\mu_1, \dots, \mu_{d-2}$.

Central transversals and mass assignments

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Theorem (Schnider 2020)

Let $0 \leq \lambda < k \leq d$ be integers. For any $(d - k) + \lambda + 1$ k -dimensional mass assignments in $\mathbb{R}^d$, there is a k -dimensional linear subspace $S_k \subset \mathbb{R}^d$ in which all the mass assignments share a λ -transversal.

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- ▶ the case $k = d$ is the central transversal theorem.
- ▶ the case $\lambda = k - 1$ is a ham sandwich for mass assignments.

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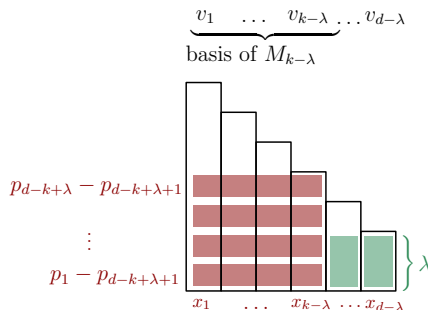
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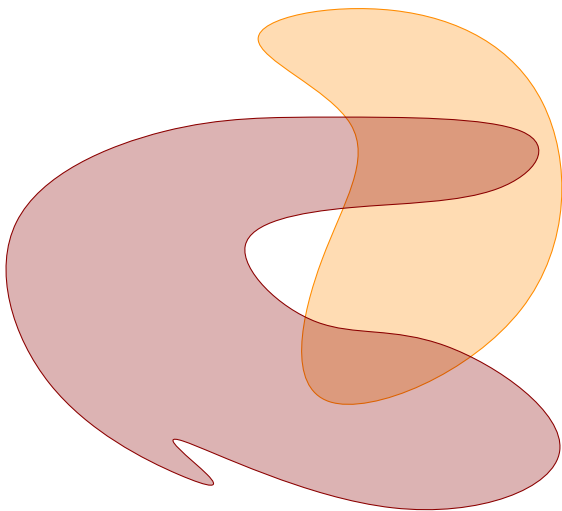
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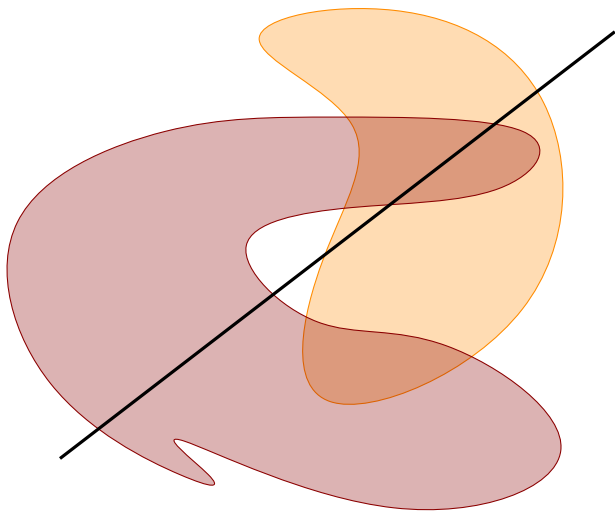
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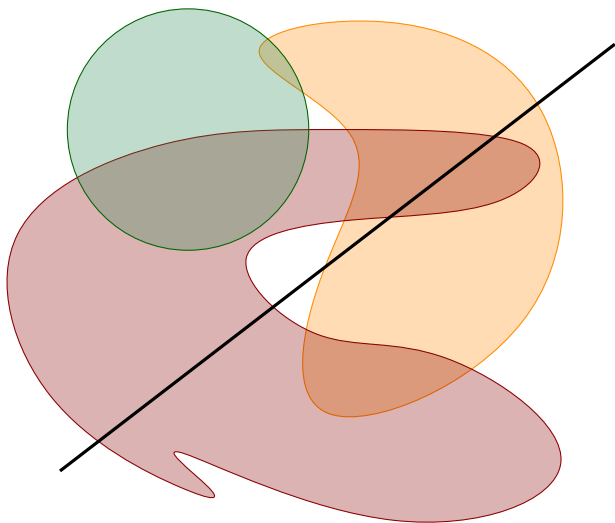
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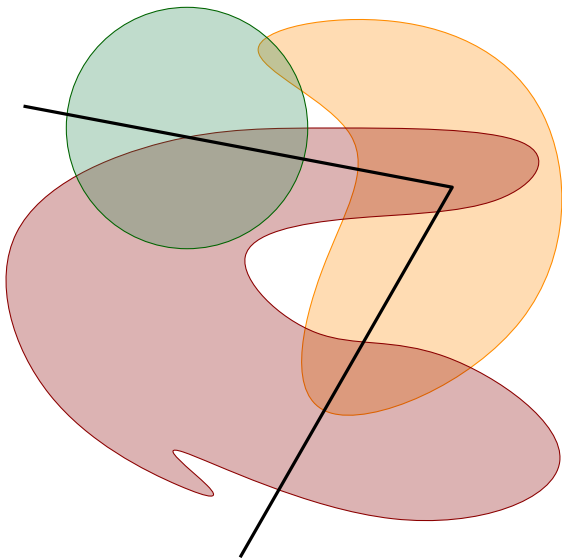


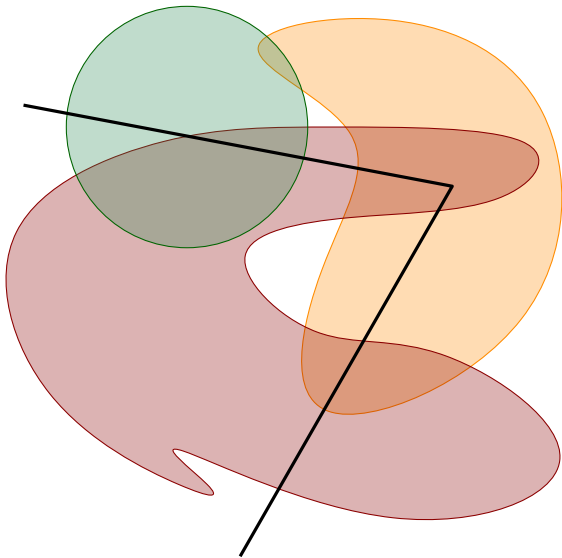
ONE MORE APPLICATION





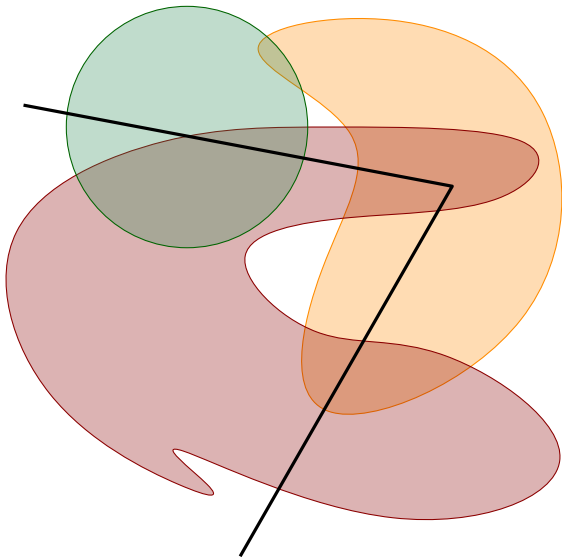






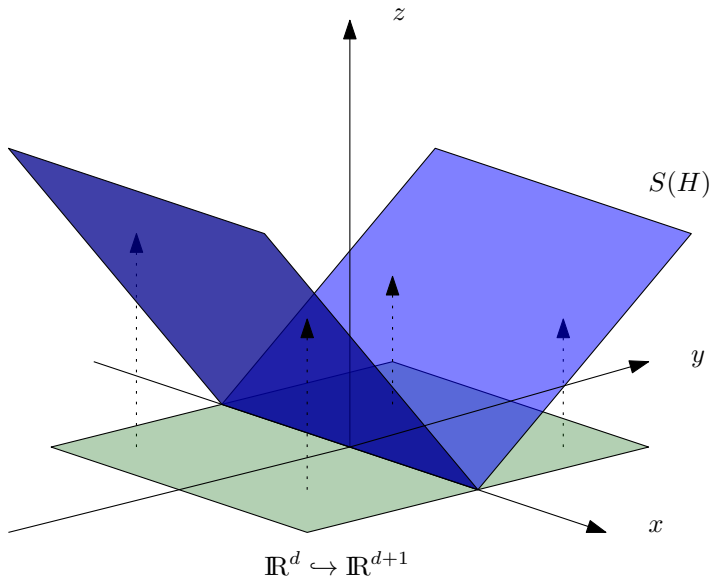
Theorem (Bárány, Matoušek 2001)

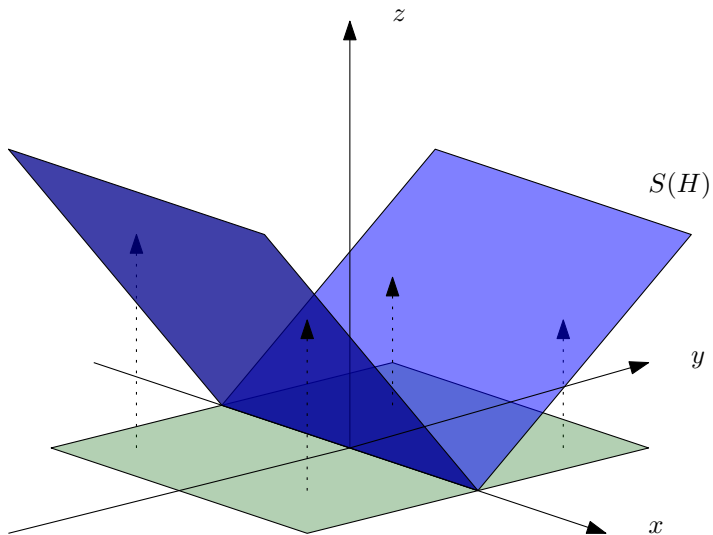
For any **3** finite measures in **the plane**, there exists a wedge containing exactly half of each measure.



Theorem (Schnider 2020+)

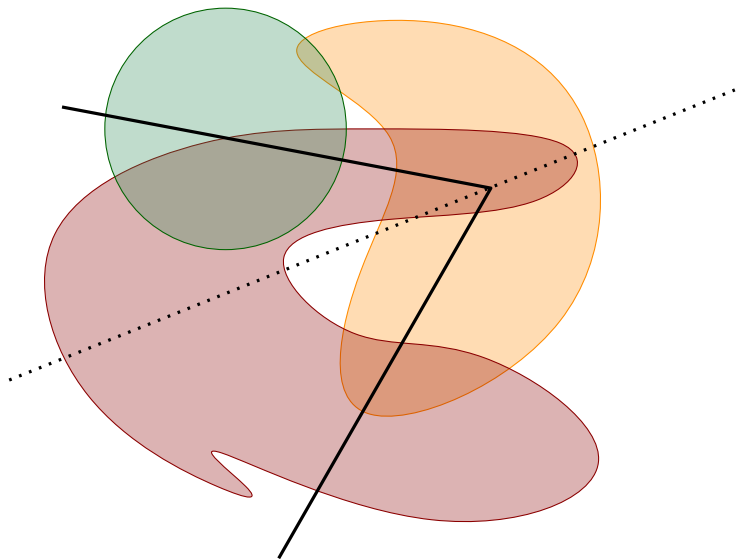
For any $d + 1$ finite measures in $\mathbb{R}^d$, there exists a wedge containing exactly half of each measure.

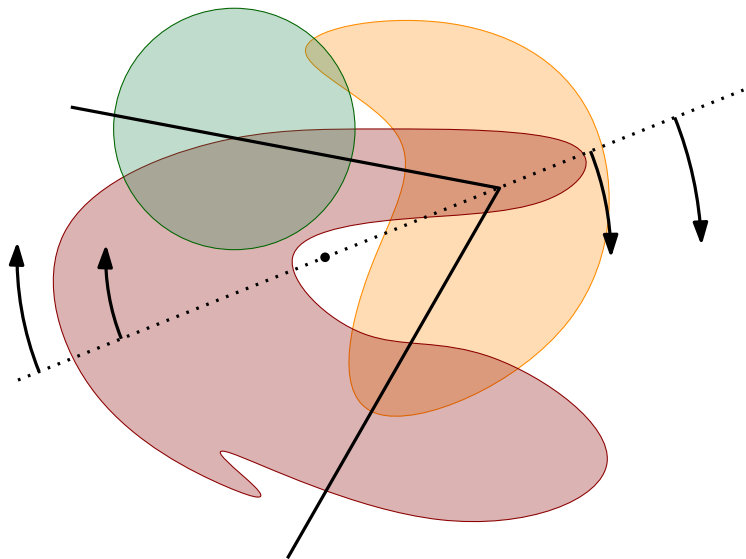


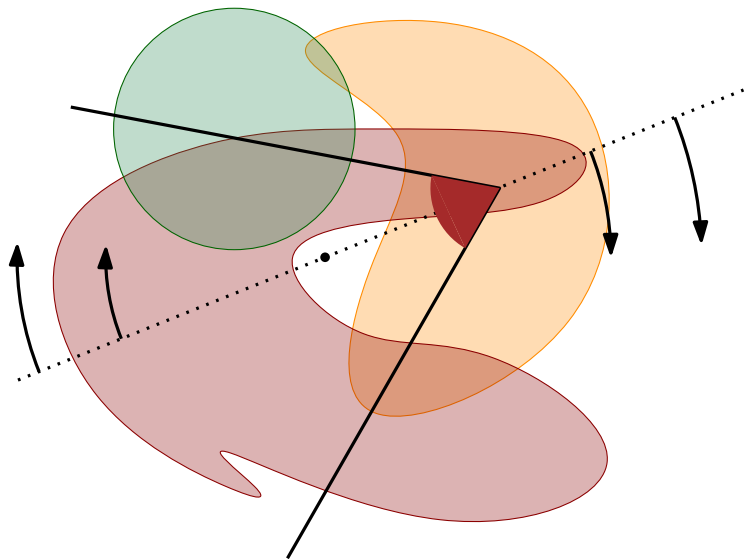


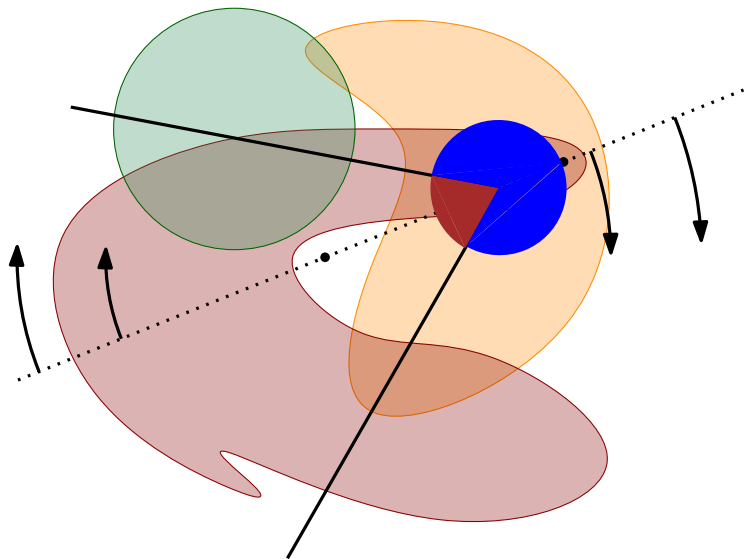
$$\mathbb{R}^d \hookrightarrow \mathbb{R}^{d+1}$$

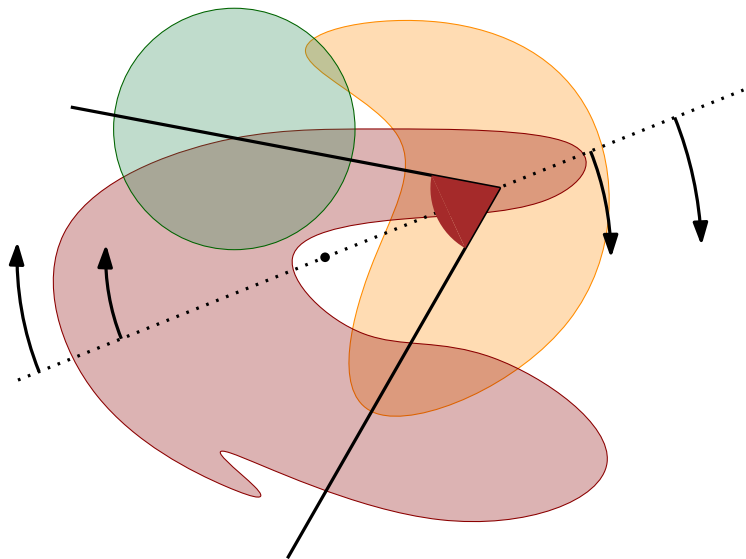
Lift the measures to $\mathbb{R}^{d+1}$, then use the ham sandwich theorem.

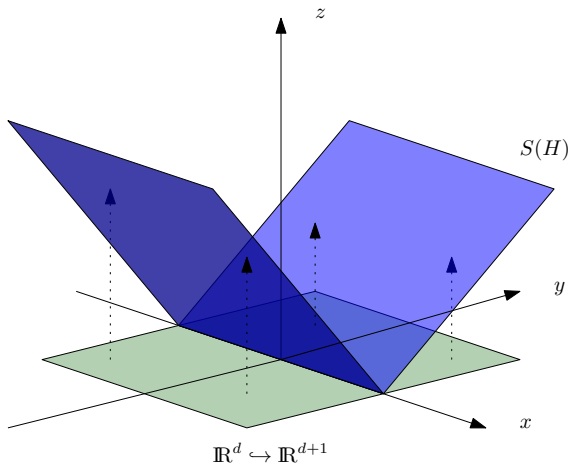


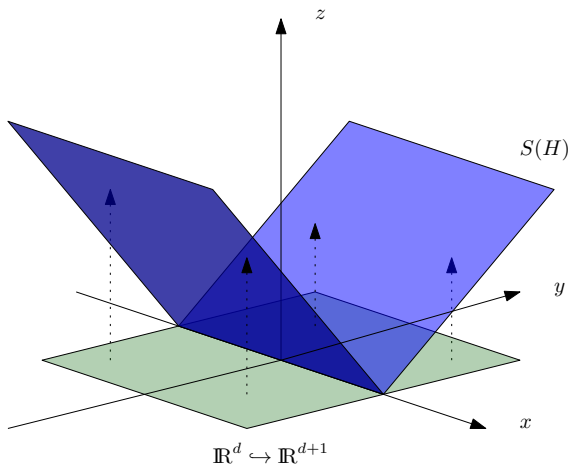




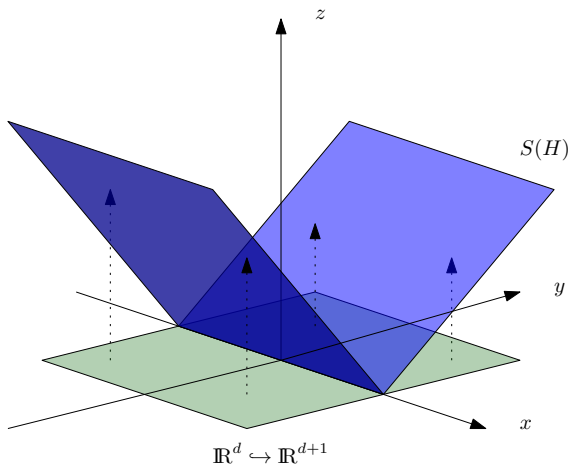




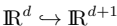




- Basis of H can be parametrized with $V_{d-1}(\mathbb{R}^d)$.



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- An oriented half-space in $\mathbb{R}^{d+1}$ halving one fixed measures can be parametrized with S^d .



- ## Question

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Lemma (Soberón, Takahashi 2021+)

*Every continuous $(\mathbb{Z}_2)^{k+1}$ -equivariant map
 $f : S^d \times V_k(\mathbb{R}^d) \rightarrow \mathbb{R}^d \times \mathbb{R}^{d-1} \times \dots \times \mathbb{R}^{d-k}$ must have a zero.*

Lemma (Soberón, Takahashi 2021+)

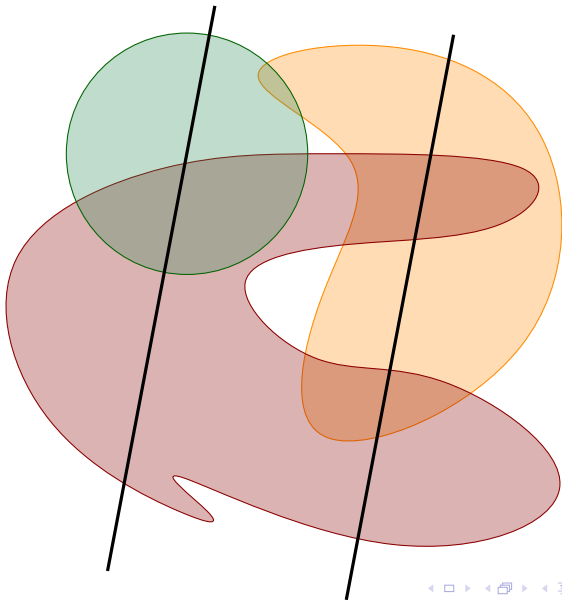
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Theorem (Soberón, Takahashi 2021+)

For any $d + 1$ measures in $\mathbb{R}^d$ we can find two parallel hyperplanes such that the region between them contains exactly half of each measure.

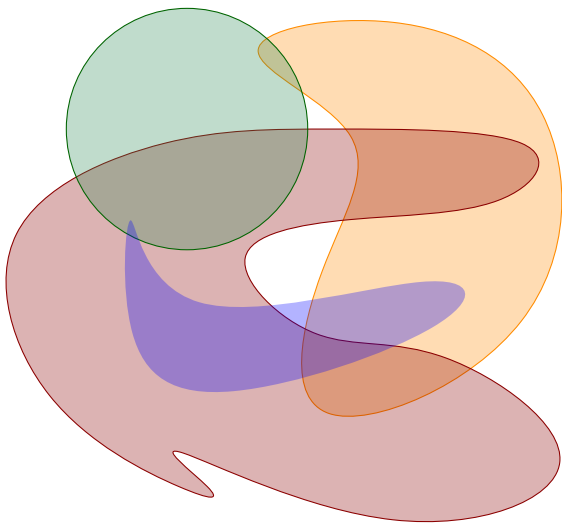
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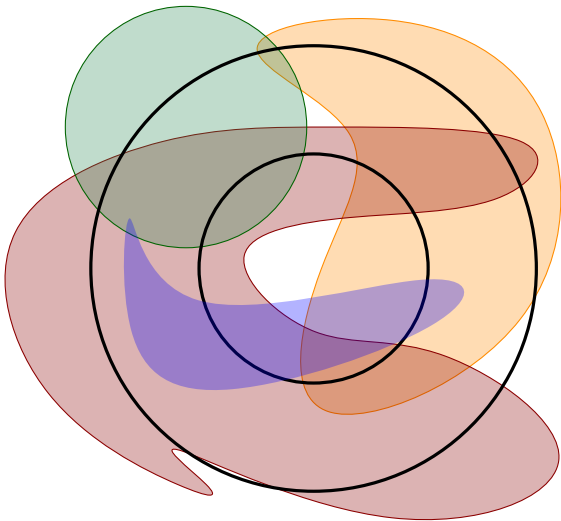
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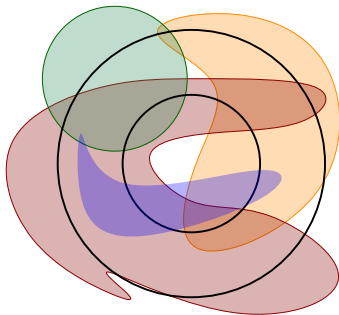
Theorem (Soberón, Takahashi 2021+)

For any $d + 2$ measures in $\mathbb{R}^d$ we can find two *concentric spheres* such that the region between them contains exactly half of each measure.



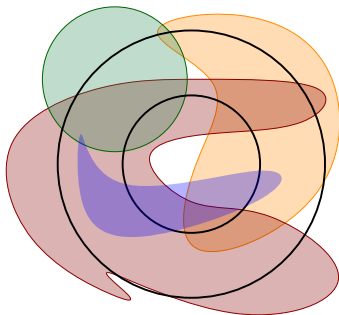
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Idea: Lift $\mathbb{R}^d$ to the paraboloid in $\mathbb{R}^{d+1}$

$$\mathcal{P} = \left\{ \left(x_1, \dots, x_d, \sum_{i=1}^d x_i^2 \right) : x_1, \dots, x_d \in \mathbb{R} \right\}.$$